

classmate

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General Solution to a large Class of Recurrence

The time requirement of recurrence based solution can be expressed as follows in general

$$T(n) = aT(n/b) + d(n) \quad \text{--- (1)}$$

$n \rightarrow$ input size

$a, b \rightarrow$ the solution is expressed as a number of recursive calls with each sub program of size n/b .

$d(n) \rightarrow$ indicates time req^d to combine the solution.

Solution for expression of eqn (1) is expressed as a summation of two sections

$$T(n) = HS + PS$$

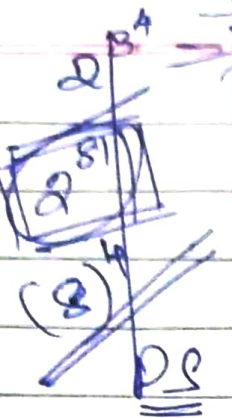
Homogeneous Solⁿ

Particular Solⁿ

$$3 \star 4 / 5$$

$$12/5 = 2.4 \quad \text{class}$$

$$4/5 = 0.8 \times 2 = 1.6$$



$$T(n) = O(n) + O(n \log n)$$

$$\underline{\underline{T(n) = O(n \log n)}}$$

$$T(n) = 4 T(n/2) + n$$

$$d(n) = n$$

$$a=4, b=2, d(b)=2$$

$$HS = O(n^{\log_b a}) = n^{\log_2 4} = O(n^2)$$

$$PS = O(n^{\log_b a}) = O(n^{\log_2 4}) = O(n^2)$$

$$\therefore T(n) = O(n^2)$$

$$T(n) = 4 T(n/2) + n^2$$

$$a=4, b=2, d(b)=n^2=2^2=4$$

$$a = d(b)$$

$$PS = \log n \cdot n^{\log_2 4} = n^2 \log n$$

$$T(n) = \underline{\underline{O(n^2 \log n)}}$$

$$6 * 5 * 10/9 - 3$$

$$300/9 - 3$$

$$6 * 5 * ((4+8) - 2) / 9 - 3$$

$$(a+b) * (c+d) = ab + cd + \dots$$